

Quantum Field Theory

We're now going to pivot to introducing quantum field theory — the theoretical framework underlying all of particle physics, which we'll use extensively in calculations for the rest of the course.

After some motivation, we'll review classical field theory and discuss symmetries from a field theory perspective.

We'll then turn to quantization of scalar fields, using canonical quantization.

We won't have time to discuss path integrals — for that you'll have to consult a real field theory class.

This section will deal almost entirely with free fields.

We'll then turn our attention to interacting fields in perturbation theory, which leads naturally to scattering processes (and back to particle physics).

But why quantum fields?

A field is just a function of spacetime — it gives a value at each point $x^\mu = (\vec{x}, t)$.

This generalizes our classical / quantum mechanics where the degree of freedom q is just a function of t .

In fact, we'll see that QFT is essentially the QM of an infinite number of d.o.f's, living in 3+1 rather than 1d.

But why is this necessary? In the particle physics context, this is ~~because~~ necessitated by special relativity.

Some (hand-wavy) arguments:

- Antiparticles : The study of relativistic wave equations (Klein-Gordon, Dirac, Maxwell) reveals the presence of "negative energy" solutions, which seem problematic in a fundamental, quantum mechanical theory.

In QFT, these are naturally associated w/ (entirely non-problematic) anti-particle states.

- Particle creation / destruction :

QFT makes exchanging energy for particles via $E = \sqrt{p^2 + m^2}$ very simple, due to the infinite d.o.f.

→ This is otherwise very cumbersome in QM.

- Causality - we do not want signals travelling faster than the speed of light.

In ordinary QM, the amplitude for a particle propagating b/w spacelike separated points is nonzero. In QFT, this amplitude is (essentially) cancelled by the antiparticle propagation. This ensures causality is preserved.

- We already have fields! - the $\vec{E}$ & $\vec{B}$ fields in a fundamental theory are dynamical so it's essential to work in a context where we quantize these. In fact, these are a much richer, more complicated type of field we'll return to later on, called a gauge field.

With that motivation, we turn to classical field theory.

Classical Lagrangian Field Theory

Recall for a single d.o.f., $q(t)$, we wrote the action S as the time integral of the Lagrangian L :

$$S = \int dt L\left(q, \frac{dq}{dt}\right) = \int dt \left[\frac{1}{2} \left(\frac{dq}{dt}\right)^2 - V(q) \right]$$

We then extremized the action,

$$\begin{aligned} \delta S = 0 &= \int dt \left[\left(\frac{dq}{dt}\right) \frac{d}{dt} \delta q - \frac{dV}{dq} \delta q \right] \\ &= - \int dt \left[\frac{d^2 q}{dt^2} + \frac{dV}{dq} \right] \delta q \quad (\text{dropping the bdy term}) \end{aligned}$$

to find the equations of motion describing the dynamics:

$$\frac{d^2 q}{dt^2} + \frac{dV}{dq} = 0.$$

Now we generalize this by replacing $q(t)$ with a scalar field $\phi(x^\mu) = \phi(t, \vec{x}) \equiv \phi(x)$.

Scalar here means that at each spacetime point, ϕ just returns a single c-number.

The integration measure must also be generalized:

$$dt \rightarrow d^4 x = dt d^3 \vec{x} = dt dx dy dz$$

and the Lagrangian becomes a functional of the scalar field and its gradients.

Since we want a relativistic theory, we'll want the (scalar) Lagrangian to be invariant under Lorentz transformations. This suggests we write everything in terms of the four-gradient:

$$\frac{d}{dt} \rightarrow \frac{\partial}{\partial x^m} \equiv \partial_m = \left(\frac{\partial}{\partial t}, \vec{\nabla} \right)$$

With these replacements, the action of the scalar field is:

$$\begin{aligned} S &= \int d^4x \mathcal{L}(\phi, \partial_m \phi) \\ &= \int d^4x \left[\frac{1}{2} g^{mn} \partial_m \phi \partial_n \phi - V(\phi) \right] \end{aligned}$$

where $\mathcal{L}$ is the Lagrangian density. (I will never remember to say density).

Note that both $\mathcal{L}$ and S are Poincaré invariant.

Now, just as for the generalized coordinate, we apply the principle of least action

— variational calculus of functions works the same as for ordinary coordinates.

(6)

We find:

$$\begin{aligned}
 0 = \delta S &= \int d^4x \left[g^{\mu\nu} \partial_\mu \phi \partial_\nu \delta\phi - \frac{\partial V}{\partial \phi} \delta\phi \right] \\
 &= - \int d^4x \delta\phi \left[g^{\mu\nu} \partial_\mu \partial_\nu \phi + \frac{\partial V}{\partial \phi} \right] \\
 &\quad + \int d^4x \partial_\nu \left[g^{\mu\nu} \partial_\mu \phi \delta\phi \right]
 \end{aligned}$$

The last term can be converted to a surface integral over the boundary of the four-volume using Stokes's Theorem.

Assuming $\delta\phi$ vanishes at $|x| \rightarrow \infty$, but is otherwise arbitrary, we obtain the eqn. of motion for the scalar field:

$$g^{\mu\nu} \partial_\mu \partial_\nu \phi + \frac{\partial V}{\partial \phi} = 0$$

$$\Rightarrow \square \phi + \frac{\partial V}{\partial \phi} = 0, \quad \text{where } \square = \partial_\mu \partial^\mu = g^{\mu\nu} \partial_\mu \partial_\nu$$

In 3-dimensional notation, this gives:

$$\frac{\partial^2 \phi}{\partial t^2} - \vec{\nabla}^2 \phi + \frac{\partial V}{\partial \phi} = 0$$

In the absence of a potential this is just a wave equation; yields the expected dynamics of an ordinary "field".

For most of this section, we'll work with the simplest allowed potential:

$$V(\phi) = \frac{1}{2} m^2 \phi^2$$

So far, m is just a parameter.

This will be the analogue of the harmonic oscillator — it is called free field theory — there are no interactions (which would come from terms $\sim \phi^3$ or higher)

We'll see soon that we are able to quantize the free theory exactly.

The equation of motion is:

$$\square \phi + m^2 \phi = 0.$$

This is known as the Klein-Gordon equation.

To interpret the parameter m , consider turning on a point source for the field (like adding a point charge in EM) — the source will ~~lead~~ set up a static $\phi(\vec{r})$ field, the solution to the equation:

$$-\nabla^2 \phi(r) + m^2 \phi(r) = -g \delta^3(\vec{r})$$

where g is a coupling constant.

The solution to this equation is:

$$\phi(\vec{r}) = \phi(r) = -\frac{g}{4\pi} \frac{1}{r} e^{-mr} \quad (r = |\vec{r}|)$$

→ The "force" associated with ϕ has a finite range.
 $R \sim m^{-1}$.

This is the familiar Yukawa potential, w/ m having the interpretation of a mass.

We'll later see this is exactly the right interpretation for many reasons.

Hamiltonian Field Theory

We'll soon quantize our field in the canonical formalism.

To do this, first we derive the Hamiltonian classically, then impose canonical commutation relations, with everything upgraded to operators.

First, define the "conjugate momentum density"

$$\pi(t, \vec{x}) = \frac{\delta \mathcal{L}}{\delta \dot{\phi}(t, \vec{x})} = \frac{\delta \mathcal{L}}{\delta \dot{\phi}(t, x)}$$

* the derivative

As always, the Hamiltonian (density) is the Legendre transform:

$$H = \int d^3x \mathcal{H} = \int d^3x [\pi(t, \vec{x}) \dot{\phi}(t, \vec{x}) - \mathcal{L}]$$

For the free theory, $\pi(t, \vec{x}) = \dot{\phi}(t, \vec{x})$, and

$$\begin{aligned} H &= \int d^3x \left[\underbrace{\pi \dot{\phi}}_{=\pi^2} - \left(\underbrace{\frac{1}{2} \dot{\phi}^2}_{=\pi^2} - \frac{1}{2} (\vec{\nabla} \phi)^2 - \frac{1}{2} m^2 \phi^2 \right) \right] \\ &= \int d^3x \left[\frac{1}{2} \pi^2 + \frac{1}{2} (\vec{\nabla} \phi)^2 + \frac{1}{2} m^2 \phi^2 \right] \end{aligned}$$

The three terms in the Hamiltonian represent the energy densities associated with moving in time, steering in space, and in potential energy, respectively.

~~Symmetries, Conservation~~

Symmetries, Conservation Laws & Noether's Theorem

In QFT, there is an intimate relationship between symmetries & conservation laws, encapsulated in a theorem by German ~~mathematician~~ ^{mathematical physicist} Emmy Noether!

Consider a continuous transformation of $\phi(x)$:

In infinitesimal form, we write

$$\phi(x) \rightarrow \phi'(x) = \phi(x) + \alpha \Delta\phi(x)$$

α inf. parameter

w/ $\Delta\phi$ some deformation of the field configuration.

This transformation is a symmetry of the theory if it leaves the equations of motion invariant.

This is true if the action remains the same under the transformation, up to a surface term.

Thus, we require the Lagrangian to be invariant up to a '4-divergence':

$$\mathcal{L}(x) \rightarrow \mathcal{L}(x) + \alpha \partial_\mu \mathcal{J}^\mu(x)$$

for some $\mathcal{J}^\mu(x)$.

Compare this to the result obtained by varying the fields:

$$\begin{aligned}
 \alpha \Delta \mathcal{L} &= \frac{\partial \mathcal{L}}{\partial \phi} (\alpha \Delta \phi) + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\mu (\alpha \Delta \phi) \\
 &= \alpha \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \Delta \phi \right] + \alpha \underbrace{\left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right]}_{=0 \text{ by eqns. of motion!}} \Delta \phi
 \end{aligned}$$

The first term can be identified as $\alpha \partial_\mu \mathcal{T}^\mu$:

$$\alpha \Delta \mathcal{L} = \alpha \partial_\mu \mathcal{T}^\mu = \alpha \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \Delta \phi \right]$$

Equivalently,

~~$$0 = \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \Delta \phi - \mathcal{T}^\mu \right]_{,\mu}$$~~

$$0 = \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \Delta \phi - \mathcal{T}^\mu \right] \equiv \partial_\mu j^\mu$$

where we've defined a 4-current (density):

$$j^\mu(x) = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \Delta \phi - \mathcal{T}^\mu$$

We see that this current is conserved $\partial_\mu j^\mu = 0$;

A continuous symmetry (specifies $\Delta \phi$, $\mathcal{T}^\mu$) leads to a conservation law.

This conservation law can be expressed in terms of a conserved charge Q , defined as:

$$Q \equiv \int_{\text{all space}} d^3x j^0(x)$$

Since, (since $\partial_\mu j^\mu = 0 = \partial_0 j^0 - \partial_i j^i$)

$$\frac{dQ}{dt} = \int d^3x \frac{dj^0(x)}{dt} = - \int \vec{\nabla} \cdot \vec{j}(x) dx$$

$$= \int_{\text{boundary of space}} \vec{j}(x) \cdot d\vec{S} = 0$$

The last term vanishes assuming the current vanishes at the spatial boundary. The total charge Q is unchanged in time.

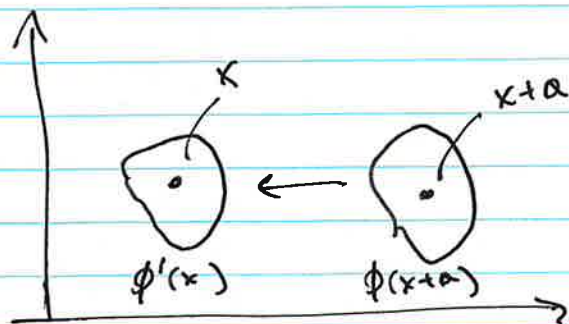
We can apply these ideas to the spacetime symmetries we just worked with.

Consider the translation: $x^\mu \rightarrow x^\mu + a^\mu$ ($a = \text{const.}$)

The scalar field is transformed as:

$$\phi(x) \rightarrow \phi'(x) = \phi(x + a) \simeq \phi(x) + \underbrace{a^\mu \partial_\mu \phi(x)}_{\equiv \Delta \phi}$$

This is an "active" transformation, in which we keep the coordinate system fixed and translate the field configuration!



The Lagrangian is also a scalar field, so it transforms in the same way!

$$\mathcal{L} \rightarrow \mathcal{L} + a^\mu \partial_\mu \mathcal{L} = \mathcal{L} + \underbrace{a^\nu \partial_\mu (\delta^\mu_\nu \mathcal{L})}_{\Delta \mathcal{L} = a^\nu \partial_\mu (T^\mu_\nu)}$$

(note our infinitesimal parameter has a spacetime index
 $\rightarrow$ the "current" will have a separate one in addition to this)

Applying Noether's theorem, we identify

$$\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \Delta \phi = a^\nu \left[\frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi)} \partial_\nu \phi \right], \quad T^\mu_\nu = \mathcal{L} \delta^\mu_\nu$$

So we get four, separately conserved currents for $\nu=0, 1, 2, 3$.

These are conveniently grouped into a tensor known as the stress-energy tensor.

$$T^{\mu}_{\nu} \equiv \frac{\delta \mathcal{L}}{\delta(\partial_{\mu}\phi)} \partial_{\nu}\phi - \mathcal{L} \delta^{\mu}_{\nu}$$

which satisfies $\partial_{\mu} T^{\mu}_{\nu} = 0$.

Note that the conserved charge associated with time translations ($\nu=0$) is the Hamiltonian!

$$\begin{aligned} H &\equiv \int T^{00} d^3x = \int T^0_{\nu} g^{\nu 0} d^3x = \int T^0_0 d^3x \\ &= \int d^3x \left[\frac{\delta \mathcal{L}}{\delta \dot{\phi}} \dot{\phi} - \mathcal{L} \right] \equiv \int d^3x \mathcal{H}. \end{aligned}$$

The conserved charges associated w/ spatial translations are

$$\begin{aligned} P^i &\equiv \int T^{0i} d^3x = - \int T^0_i d^3x \\ &= - \int \frac{\delta \mathcal{L}}{\delta \dot{\phi}} \partial_i \phi d^3x = - \int d^3x \pi \partial_i \phi \end{aligned}$$

This can be interpreted as the physical momentum carried by the field.

We see that the spacetime symmetries of translations lead to energy & momentum conservation, respectively.

Besides spacetime symmetries, a theory may possess internal symmetries (ones not related to a change of space-time coordinates).

Consider a complex scalar field with Lagrangian:

$$\begin{aligned} \mathcal{L} &= |\partial_\mu \phi|^2 - m^2 |\phi|^2 \\ &= g^{\mu\nu} \partial_\mu \phi^* \partial_\nu \phi - m \phi^* \phi \end{aligned}$$

(Note the lack of $1/2$ factors — this is just convention for now, but you should think of it as counting the two separate "real" fields, $\phi = \phi_1 + i\phi_2$.)

This Lagrangian is invariant under the transformation

$$\phi(x) \rightarrow e^{i\alpha} \phi(x) \quad \text{for } \alpha = \text{const.}$$

This is a global $U(1)$ symmetry.

It is "global" in that α does not depend on x^μ , as opposed to a "local" transformation by $\alpha(x^\mu)$.

The $U(1)$ refers to the Lie group associated w/ this symmetry. A $U(1)$ symmetry is an Abelian symmetry group associated w/ rephasing of fields.

One can show that the conserved current in this theory is:

$$j^\mu(x) = i [(\partial^\mu \phi)^* \phi - \phi^* \partial^\mu \phi]$$

We'll have more to say about this later on.

Canonical Quantization

Shortly, we will proceed w/ the canonical quantization of the free scalar field.

As a brief warm-up, let's remind ourselves how the simple harmonic oscillator works in ordinary QM. The Hamiltonian is

$$H = \frac{1}{2m} \hat{P}^2 + \frac{1}{2} m \omega^2 \hat{X}^2$$

w/ $\hat{X}, \hat{P}$ the operators satisfying $[\hat{X}, \hat{P}] = i\hbar$, m is the particle mass, ω the oscillator frequency.

We can introduce the "ladder operators", $a, a^\dagger$ in terms of X, P :

$$\hat{a} = \left(\frac{m\omega}{2\hbar}\right)^{1/2} \hat{X} + i \left(\frac{1}{2m\omega\hbar}\right)^{1/2} \hat{P}$$

$$\hat{a}^\dagger = \left(\frac{m\omega}{2\hbar}\right)^{1/2} \hat{X} - i \left(\frac{1}{2m\omega\hbar}\right)^{1/2} \hat{P}$$

(recall X, P are Hermitian)

One can work out that these satisfy the commutation relation

$$[\hat{a}, \hat{a}^\dagger] = 1.$$

Check:

$$\begin{aligned}
 [a, a^\dagger] &= \left[\left(\frac{m\omega}{2\hbar} \right)^{1/2} \hat{X} + i \left(\frac{1}{2m\omega\hbar} \right)^{1/2} \hat{P}, \left(\frac{m\omega}{2\hbar} \right)^{1/2} \hat{X} - i \left(\frac{1}{2m\omega\hbar} \right)^{1/2} \hat{P} \right] \\
 &= \left(\frac{m\omega}{2\hbar} \frac{1}{2m\omega\hbar} \right)^{1/2} \left(-i[X, P] + i[P, X] \right) \\
 &= -[X, P] \\
 &= \frac{1}{2\hbar} (-i)(2\hbar)(i\hbar) = 1.
 \end{aligned}$$

The utility of this stems from the fact that the Hamiltonian is related to $a^\dagger a$:

$$\begin{aligned}
 a^\dagger a &= \left[\left(\frac{m\omega}{2\hbar} \right)^{1/2} \hat{X} - i \left(\frac{1}{2m\omega\hbar} \right)^{1/2} \hat{P} \right] \left[\left(\frac{m\omega}{2\hbar} \right)^{1/2} \hat{X} + i \left(\frac{1}{2m\omega\hbar} \right)^{1/2} \hat{P} \right] \\
 &= \frac{m\omega}{2\hbar} \hat{X}^2 + \frac{1}{2m\omega\hbar} \hat{P}^2 + \underbrace{\frac{i}{2\hbar} [X, P]}_{=i\hbar} \\
 &= \frac{1}{\omega\hbar} \left[\hat{P}^2 \cdot \frac{1}{2m} + \frac{1}{2} m\omega^2 \hat{X}^2 \right] - \frac{1}{2} \\
 &= \frac{1}{\omega\hbar} \hat{H} - \frac{1}{2}
 \end{aligned}$$

$$\text{or, } \hat{H} = \hbar\omega \left(a^\dagger a + \frac{1}{2} \right)$$

The ladder operators are also useful in diagonalizing H . Consider the rescaled operator $H' = H/\hbar\omega$ (purely for convenience - it measures energy in units of $\hbar\omega$)

Now we'll solve the eigenvalue problem for H' :

$$H'|\epsilon\rangle = \epsilon|\epsilon\rangle$$

ϵ is the energy eigenvalue (in units of $\hbar\omega$).

We'll use the relations:

$$[a, H'] = a, \quad [a^\dagger, H'] = -a^\dagger$$

which can easily be checked:

$$\begin{aligned} [a, H'] &= [a, a^\dagger a] = a a^\dagger a - a^\dagger a a \\ &= [a, a^\dagger] a = 1 \cdot a = a \end{aligned}$$

$$\begin{aligned} [a^\dagger, H'] &= [a^\dagger, a^\dagger a] = a^\dagger a^\dagger a - a^\dagger a a^\dagger \\ &= a^\dagger [a^\dagger, a] = -1 \cdot a^\dagger = -a^\dagger \end{aligned}$$

These relations tell us that when $a, a^\dagger$ act on an eigenstate, they return a different eigenstate:

starting w/ an eigenstate $|\epsilon\rangle$, we check if $a|\epsilon\rangle$ is an eigenstate of H' :

$$\begin{aligned} H'(a|\epsilon\rangle) &= [H'a - aH' + aH']|\epsilon\rangle \\ &= (-[a, H'] + \epsilon a)|\epsilon\rangle = (-a + \epsilon a)|\epsilon\rangle \\ &= (\epsilon - 1)(a|\epsilon\rangle) \end{aligned}$$

We see that $a|z\rangle$ is an eigenstate of H' w/ eigenvalue $z-1$.

$$a|z\rangle = c_z |z-1\rangle$$

$\uparrow$ const. to be determined.

$\rightarrow |z\rangle, |z-1\rangle$ are normalized eigenvectors of H' .

a lowers the energy by 1 unit, known as a "lowering" operator.

Similarly, we can consider $a^\dagger$ acting on $|z\rangle$:

$$H'(a^\dagger|z\rangle) = [H'a^\dagger - a^\dagger H' + a^\dagger H']|z\rangle$$

$$= (-[a^\dagger, H'] + z a^\dagger)|z\rangle$$

$$= (z+1)a^\dagger|z\rangle \quad \leftarrow \text{eigenvector w/ eigenvalue } z+1!$$

So

$$a^\dagger|z\rangle = c_{z+1}|z+1\rangle$$

and we refer to $a^\dagger$ as a "raising operator".

We will more frequently refer to these as creation/annihilation operators ($a^\dagger$ & a), since they create/destroy quanta of energy.

$\Rightarrow$ starting w/ some z , we get eigenstates w/ energies $z+1, z+2, \dots$ and $z-1, z-2, \dots$ down to a minimum, positive value.

(The Hamiltonian is positive definite, so there can't be any negative energy eigenstates)

There is thus a state $|z_0\rangle$ which cannot be lowered further:

$$a|z_0\rangle = 0$$

Then, acting w/ $a^\dagger$ we find:

$$a^\dagger a|z_0\rangle = 0$$

$$= (H' - \frac{1}{2})|z_0\rangle = 0 \Rightarrow H'|z_0\rangle = \frac{1}{2}|z_0\rangle$$

So this is the ground state: $z_0 = 1/2$.

The rest of the energy levels are given by

$$z_n = n + 1/2, \text{ or } E_n = (n + 1/2)\hbar\omega$$

$$n = 0, 1, 2, \dots$$

For completeness, we derive the coefficients introduced by $a, a^\dagger$. From here on we denote each eigenket $|n\rangle$. Then,

$$a|n\rangle = c_n|n-1\rangle$$

taking the adjoint:

$$\langle n|a^\dagger = \langle n-1|c_n^*$$

$$\Rightarrow \langle n|a^\dagger a|n\rangle = \underbrace{\langle n-1|n-1\rangle}_{=1} |c_n|^2$$

$$\Rightarrow \langle n|H' - \frac{1}{2}|n\rangle = |c_n|^2 = \langle n|(n + \frac{1}{2} - \frac{1}{2})|n\rangle$$

So

$$|c_n|^2 = n \langle n | n \rangle = n$$

$$\hookrightarrow \text{ ~~} c_n = \sqrt{n} \text{ } \text{ } c_n = (n)^{1/2}~~$$

Similarly, from $a^\dagger |n\rangle = c_{n+1} |n+1\rangle$ we can show

$$c_{n+1} = (n+1)^{1/2}$$

So in summary:

$$a |n\rangle = (n)^{1/2} |n-1\rangle$$

$$a^\dagger |n\rangle = (n+1)^{1/2} |n+1\rangle$$

Now consider the operator $N = a^\dagger a$ acting on $|n\rangle$:

$$\begin{aligned} \hat{N} |n\rangle &= a^\dagger a |n\rangle = (n)^{1/2} a^\dagger |n-1\rangle \\ &= (n)^{1/2} (n-1+1)^{1/2} |n\rangle = n |n\rangle \end{aligned}$$

So $\hat{N}$ returns the integer n , and thus counts the number of quanta. It is called the "number operator".

Using $a^\dagger |n\rangle = (n+1)^{1/2} |n+1\rangle$ we can express any normalized energy eigenket in terms of the ground state $|0\rangle$:

$$\begin{aligned} |n\rangle &= \frac{a^\dagger}{n^{1/2}} |n-1\rangle = \frac{a^\dagger}{n^{1/2}} \frac{a^\dagger}{(n-1)^{1/2}} |n-2\rangle = \dots \\ &= \frac{(a^\dagger)^n}{(n!)^{1/2}} |0\rangle \end{aligned}$$

Then, given any initial state ket $|\Psi(0)\rangle$, the complete solution to the Schrödinger equation is:

$$\begin{aligned}
 |\Psi(t)\rangle &= \sum_{n=0} |\psi_n\rangle \langle n | \Psi(0) \rangle e^{-iE_n t / \hbar} \\
 &= \left[\sum_{n=0} |\psi_n\rangle \langle n | e^{-i(n+\frac{1}{2})\omega t} \right] |\Psi(0)\rangle
 \end{aligned}$$

As a final note, we note that the Hamiltonian for the SHO can be derived from the Lagrangian ($m, p = m\dot{q}$)

$$L_{\text{SHO}} = \frac{1}{2} m \dot{q}^2 - \frac{1}{2} m \omega^2 q^2$$

which yields the e.o.m:

$$0 = \frac{d}{dt} \left(\frac{\delta L}{\delta \dot{q}} \right) - \frac{\delta L}{\delta q} = m \ddot{q} + m \omega^2 q$$

$$\text{or simply, } \ddot{q} + \omega^2 q = 0.$$

Now we return to quantum fields.

Field Quantization

With the QM analogy in hand, we proceed to quantize our real scalar free field theory.

The basic point will be that we can identify the scalar as an infinite set of harmonic oscillators. To see this, consider the Fourier transform of the scalar field:

$$\phi(t, \vec{x}) = \int \frac{d^3 \vec{k}}{(2\pi)^3} e^{i\vec{k} \cdot \vec{x}} \phi(t, \vec{k})$$

Here, $\vec{k}$ is the "wavenumber", which has units of inverse length — in natural units, however, we can identify this w/ the momentum ~~of the field~~

$$\vec{k} \simeq \vec{p}$$

Plugging this into the Klein-Gordon equation,

$$\left(\frac{\partial^2}{\partial t^2} - \vec{\nabla}^2 + m^2 \right) \int \frac{d^3 \vec{p}}{(2\pi)^3} e^{i\vec{p} \cdot \vec{x}} \phi(t, \vec{p}) = 0$$

$$\Rightarrow \left[\frac{\partial^2}{\partial t^2} + (|\vec{p}|^2 + m^2) \right] \phi(t, \vec{p}) = 0$$

This is the same equation of motion as that of a Harmonic Oscillator, w/ frequency:

$$\omega_{\vec{p}} = \sqrt{|\vec{p}|^2 + m^2}$$

Thus for each momentum $\vec{p}$, $\phi(t, \vec{p})$ is on SHO w/ frequency $\omega_{\vec{p}}$.

To quantize the system, we first promote ϕ & π to quantum operators.

We'll work in the Schrödinger picture, in which ϕ, π are fixed in time:

$$\phi = \phi(\vec{x}), \quad \pi = \pi(\vec{x})$$

and impose canonical commutation relations:

$$[\phi(\vec{x}), \pi(\vec{y})] = i\delta^{(3)}(\vec{x} - \vec{y})$$

$$[\phi(\vec{x}), \phi(\vec{y})] = 0, \quad [\pi(\vec{x}), \pi(\vec{y})] = 0$$

(These are just analogues of the relations for X, P w/ $\hbar=1$)

Now, following our QM example, we write each Fourier mode in terms of creation/annihilation operators corresp. to the momentum: $a_{\vec{p}}, a_{\vec{p}}^\dagger$:

$$\phi(\vec{x}) = \int \frac{d^3\vec{p}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\vec{p}}}} (a_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} + a_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}})$$

$$\pi(\vec{x}) = \int \frac{d^3\vec{p}}{(2\pi)^3} (-i) \sqrt{\frac{\omega_{\vec{p}}}{2}} (a_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} - a_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}})$$

Note that these satisfy $(\phi)^+ = \phi$, $(\bar{\alpha})^+ = \bar{\alpha}$, since ϕ is a real scalar field.

To emphasize: there are an infinite collection of ladder operators, labeled by the continuous index $\vec{p}$.

These satisfy the commutation relations:

$$[a_{\vec{p}}, a_{\vec{p}'}^+] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{p}')$$

$$[a_{\vec{p}}, a_{\vec{p}'}] = [a_{\vec{p}}^+, a_{\vec{p}'}^+] = 0$$

One can verify explicitly that these are consistent w/ the canonical commutation relations for the field operators given above.

As in the SHO, the Hamiltonian can be written in terms of ladder operators:

$$\begin{aligned} H &= \int d^3x \mathcal{H} = \int d^3x \left[\frac{1}{2} \bar{\alpha}^2 + \frac{1}{2} (\vec{\nabla} \phi)^2 + \frac{1}{2} m^2 \phi^2 \right] \\ &= \int \frac{d^3p}{(2\pi)^3} \sum_{\vec{p}} \left(a_{\vec{p}}^+ a_{\vec{p}} + \frac{1}{2} (2\pi)^3 \delta^{(3)}(0) \right) \end{aligned}$$

(Here we've used the Fourier expansion of the δ -fcn.)

This is reminiscent of the QM example, except that we now integrate over all the modes.

The second term contains two infinities. The first is associated w/ the infinite volume of space.

$$(2\pi)^3 \delta^{(3)}(0) = \lim_{L \rightarrow \infty} \int_{-L/2}^{L/2} d^3x e^{i\vec{x} \cdot \vec{p}} \Big|_{\vec{p}=0} \rightarrow L^3 = V$$

That is, the Hamiltonian computes the total energy stored in the field, rather than the energy density. We can simply divide by the volume to remedy this.

$$\rho = \frac{H_0}{V} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2} \omega_{\vec{p}} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2} \sqrt{|\vec{p}|^2 + m^2}$$

This is still infinite — it corresponds to the sum over the zero point energies of each oscillator.

Formally, we say this expression is UV divergent.

It becomes infinite due to the high momentum modes ($p \rightarrow \infty$) contributing to the integral.

Particle physics only measures energy differences, so this zero point energy can be ignored, but this zero point energy is important for gravity which couples to all sources of energy & momentum.

In that context, this divergence is associated w/ one of the biggest outstanding problems in fundamental physics: the cosmological constant problem.

To make this more concrete, we currently understand our universe to be in a phase of accelerated expansion, driven by a vacuum energy (cosmological constant)

$$\rho_{\text{vac}} \simeq 10^{-47} \text{ GeV}^4 \simeq (10^{-3} \text{ eV})^4$$

In contrast, the ρ_{vac} we computed above for the scalar is infinite.

One way to regulate this divergence is to introduce a cutoff in momentum at some higher scale — this isn't crazy: we don't know if QFT is the appropriate description of nature at arbitrary high energies (in fact, we are reasonably sure it is not by the time we reach the Planck scale $\sim 10^{19} \text{ GeV}$)

But even if we very conservatively cut off the d^3p integral as low as $\sim \text{TeV}$, we get the value of ρ_{vac} wrong by ~ 60 orders of magnitude!

Unfortunately, that's about all I can say on this topic!

Now let's work out the spectrum of the theory.

One can show that the ladder operators & Hamiltonian satisfy:

$$[H, a_{\vec{p}}^\dagger] = \omega_{\vec{p}} a_{\vec{p}}^\dagger$$

$$[H, a_{\vec{p}}] = -\omega_{\vec{p}} a_{\vec{p}}$$

(these are analogues of the usual SHO, w/ the δ -function in momentum space)

Now, we begin with the vacuum (ground state) which has the property:

$$a_{\vec{p}} |0\rangle = 0 \quad \text{for all } \vec{p}.$$

After dropping the divergent zero-point energy, the vacuum has $E=0$.

$$H|0\rangle = \left[\int \frac{d^3p}{(2\pi)^3} \omega_{\vec{p}} a_{\vec{p}}^\dagger a_{\vec{p}} \right] |0\rangle = 0|0\rangle$$

All other eigenstates of the Hamiltonian are constructed by acting on $|0\rangle$ w/ creation operators!

In general, the state

$$a_{\vec{p}}^\dagger a_{\vec{q}}^\dagger \dots |0\rangle$$

is an eigenstate w/ energy $E = \omega_{\vec{p}} + \omega_{\vec{q}} + \dots$

Explicitly: consider

$$a_{\vec{p}}^{\dagger} |0\rangle$$

$$\begin{aligned} H(a_{\vec{p}}^{\dagger} |0\rangle) &= (H a_{\vec{p}}^{\dagger} - a_{\vec{p}}^{\dagger} H + a_{\vec{p}}^{\dagger} H) |0\rangle \\ &= [H, a_{\vec{p}}^{\dagger}] |0\rangle + a_{\vec{p}}^{\dagger} (H |0\rangle) \\ &= \omega_{\vec{p}} (a_{\vec{p}}^{\dagger} |0\rangle) \rightarrow E_{\vec{p}} = \omega_{\vec{p}}. \end{aligned}$$

One can similarly construct the spatial momentum operator.

$$\vec{P} = - \int d^3x \pi(\vec{x}) \vec{\nabla} \phi(x) = \int \frac{d^3p}{(2\pi)^3} \vec{p} a_{\vec{p}}^{\dagger} a_{\vec{p}}$$

Acting $\vec{P}$ on the state $a_{\vec{p}}^{\dagger} a_{\vec{q}}^{\dagger} \dots |0\rangle$ returns $\vec{p} + \vec{q} + \dots$ as the eigenvalue.

One can see this explicitly from:

$$\vec{P}(a_{\vec{p}}^{\dagger} |0\rangle) = \vec{p}(a_{\vec{p}}^{\dagger} |0\rangle)$$

In summary, the action of $a_{\vec{p}}^{\dagger}$ on $|0\rangle$ creates a state w/ energy $\omega_{\vec{p}} = \sqrt{|\vec{p}|^2 + m^2}$ and momentum $\vec{p}$. Acting multiple times creates a state w/ the sum of these energy/momenta.

It is natural to associate these states with particles.

This is in fact the desired outcome'. The field operator $\phi(\vec{x})$ contains a spectrum of creation operators that create particle states from the vacuum.

These states have exactly the transformation properties we expect from the Poincaré algebra (and agree w/ the algebraic derivation of 1-, 2-, ... particle states derived in the Appendix)

→ This is one of the essential utilities of QFT!

We can further study the quantum statistics of these particles associated w/ ϕ .

Consider the states $a_{\vec{p}}^{\dagger} a_{\vec{q}}^{\dagger} |0\rangle$ — since $a_{\vec{p}}^{\dagger}, a_{\vec{q}}^{\dagger}$ commute (for $\vec{q} \neq \vec{p}$), this is identical to the state $a_{\vec{q}}^{\dagger} a_{\vec{p}}^{\dagger} |0\rangle$ in which the particles are interchanged.

Furthermore, a single momentum mode can contain arbitrary many particles, e.g., the state!

$$(a_{\vec{p}}^{\dagger})^n |0\rangle$$

→ The particles associated w/ ϕ have Bose-Einstein statistics, and are an example of a boson.

Now we turn to the normalization of these states.
We choose the normalization of the vacuum such that

$$\langle 0|0\rangle = 1.$$

For the one particle states $|\vec{p}\rangle \propto a_{\vec{p}}^\dagger |0\rangle$, a common (natural) normalization is

$$\langle \vec{p} | \vec{q} \rangle = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q})$$

However, this convention is not Lorentz invariant!
Consider a boost in the z -direction:

$$E' = \gamma(E + \beta p_3), \quad p_3' = \gamma(p_3 + \beta E)$$

The δ -function satisfies the identity:

$$\delta(f(x) - f(x_0)) = \frac{1}{|f'(x_0)|} \delta(x - x_0),$$

so under this boost,

$$\delta^3(\vec{p} - \vec{q}) = \delta(p_1 - q_1) \delta(p_2 - q_2) \delta(p_3 - q_3)$$

$$\rightarrow p_1'(p_1), p_2'(p_2) = p_1, p_2$$

$$\text{but } p_3'(p_3) = \gamma(p_3 + \beta E_{\vec{p}})$$

so

$$\delta(p_3' - q_3') = \delta(\gamma(p_3 + \beta E_{\vec{p}}) - (\gamma(q_3 + \beta E_{\vec{q}})))$$

$$= \frac{1}{\gamma(1 + \beta p_3/E)} \delta(p_3 - q_3) = \frac{E}{\gamma(E + \beta p_3)} \delta(p_3 - q_3)$$

or,

$$\delta(p_3 - q_3) \rightarrow \delta(p'_3 - q'_3) = \frac{E}{E'} \delta(p_3 - q_3)$$

Thus, while the normalization $\langle \vec{p} | \vec{q} \rangle = (2\pi)^3 \delta^3(\vec{p} - \vec{q})$ is not Lorentz invariant, we see under a boost that

$$E_{\vec{p}} \delta^3(\vec{p}' - \vec{q}') = E_{\vec{p}} \delta^3(\vec{p} - \vec{q})$$

This is in fact invariant under general Lorentz transformations, so we will define

$$|\vec{p}\rangle = \sqrt{2E_{\vec{p}}} a_{\vec{p}}^\dagger |0\rangle$$

as the one-particle states in our theory.

Then,

$$\begin{aligned} \langle \vec{p} | \vec{q} \rangle &= \sqrt{2E_{\vec{p}}} \sqrt{2E_{\vec{q}}} \langle 0 | a_{\vec{p}} a_{\vec{q}}^\dagger | 0 \rangle \\ &\quad - a_{\vec{q}}^\dagger a_{\vec{p}} + [a_{\vec{p}}, a_{\vec{q}}^\dagger] \\ &= 2E_{\vec{p}} (2\pi)^3 \delta^3(\vec{p} - \vec{q}) \end{aligned}$$

(The factor 2 is for convenience, and agrees w/ the mode expansions for ϕ , $\bar{\psi}$)

Armed w/ the 1-particle states, we can study their properties a bit. As always, a Lorentz transformation is realized on the states by a unitary operator.

$$U(\Lambda) |\vec{p}\rangle = |\Lambda\vec{p}\rangle$$

$$\Rightarrow U(\Lambda) \sqrt{2E_{\vec{p}}} a_{\vec{p}}^\dagger |0\rangle = \sqrt{2E_{\Lambda\vec{p}}} a_{\Lambda\vec{p}}^\dagger |0\rangle$$

Inserting $U^{-1}(\Lambda)U(\Lambda)=1$, we see:

$$\begin{aligned} \sqrt{2E_{\vec{p}}} U(\Lambda) a_{\vec{p}}^\dagger U^{-1}(\Lambda) (U(\Lambda)|0\rangle) \\ = \sqrt{2E_{\Lambda\vec{p}}} a_{\Lambda\vec{p}}^\dagger |0\rangle \end{aligned}$$

or
the vacuum ought to
be Lorentz invariant,
so $U(\Lambda)|0\rangle = |0\rangle$

$$\Rightarrow U(\Lambda) a_{\vec{p}}^\dagger U^{-1}(\Lambda) = \sqrt{\frac{E_{\Lambda\vec{p}}}{E_{\vec{p}}}} a_{\Lambda\vec{p}}^\dagger$$

This defines how the creation/annihilation operators transform under L.T.s.

Note that we have to be careful everywhere about our conventions for 1-particle states.

E.g., the completeness relation for 1-particle states is:

$$(\mathbb{1})_{1\text{-particle}} = \int \frac{d^3p}{(2\pi)^3} |\vec{p}\rangle \frac{1}{2E_{\vec{p}}} \langle \vec{p}|$$

$\mathbb{P}$

identity operator within the subspace of 1-particle states;
evaluates to zero on the rest of Hilbert space.

Now, note that the integral:

$$\int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} = \int \frac{d^4 p}{(2\pi)^4} (2\pi) \delta(p^2 - m^2) \Big|_{p^0 > 0}$$

→ both sides are Lorentz invariant.

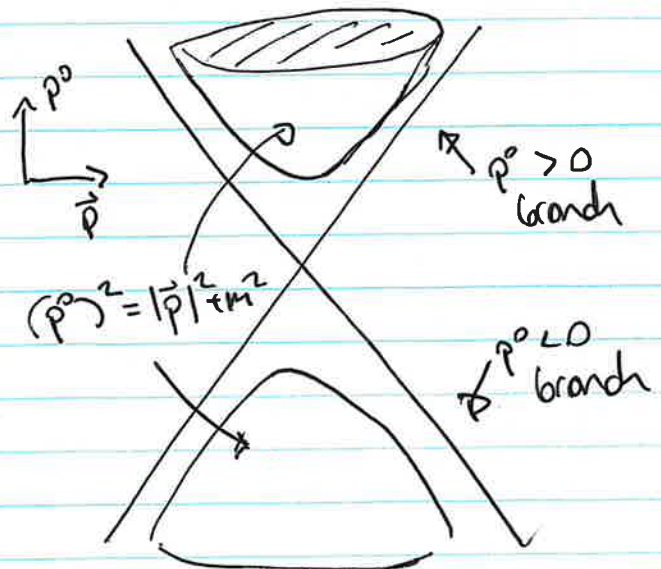
To prove this equality, note that

$$\int d p^0 \delta[(p^0)^2 - (\underbrace{|\vec{p}|^2}_{=E_{\vec{p}}^2} + m^2)] \Big|_{p^0 > 0}$$

$$= \int d p^0 \frac{1}{2p^0} \delta(p^0 - E_{\vec{p}}) \Big|_{p^0 > 0} = \frac{1}{2E_{\vec{p}}}$$

If a function $f(p)$ is Lorentz invariant, the integral $\int d^3 \vec{p} f(p) / 2E_{\vec{p}}$ is also invariant. The notation on the right makes this manifest, and will be useful later on.

The integral can be thought of as being over the positive branch of the hyperboloid $p^2 = m^2$ in momentum space.



Next, consider the state formed by the field operator:
 $\phi(\vec{x})|0\rangle$: from the mode expansion:

$$\begin{aligned}\phi(\vec{x})|0\rangle &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \left(e^{i\vec{p}\cdot\vec{x}} a_{\vec{p}}|0\rangle + e^{-i\vec{p}\cdot\vec{x}} a_{\vec{p}}^\dagger|0\rangle \right) \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} e^{-i\vec{p}\cdot\vec{x}} \sqrt{2E_{\vec{p}}} a_{\vec{p}}^\dagger|0\rangle \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} e^{-i\vec{p}\cdot\vec{x}} |\vec{p}\rangle\end{aligned}$$

$\Rightarrow \phi(\vec{x})|0\rangle$ is a linear superposition of single-particle states $|\vec{p}\rangle$ w/ well-defined momenta. Up to the factor $1/\sqrt{2E_{\vec{p}}}$, this is identical to the non-relativistic expression for an eigenstate of position $|x\rangle$, in momentum space.

We can thus interpret the field operator $\phi(x)$ as "creating a particle at x ". To further this interpretation, compute:

$$\langle 0|\phi(\vec{x})|\vec{p}\rangle = \langle 0|\int \frac{d^3p'}{(2\pi)^3} (a_{\vec{p}'} e^{i\vec{p}'\cdot\vec{x}} + a_{\vec{p}'}^\dagger e^{-i\vec{p}'\cdot\vec{x}}) \sqrt{2E_{\vec{p}'}} a_{\vec{p}}^\dagger|0\rangle$$

$$\langle 0|a_{\vec{p}'}^\dagger = 0, \quad a_{\vec{p}'} a_{\vec{p}}^\dagger|0\rangle = ([a_{\vec{p}'}, a_{\vec{p}}^\dagger] - a_{\vec{p}}^\dagger a_{\vec{p}'})|0\rangle$$

$$\begin{aligned}\hookrightarrow \langle 0|\phi(\vec{x})|\vec{p}\rangle &= \langle 0|\int \frac{d^3p'}{(2\pi)^3} \sqrt{\frac{E_{\vec{p}'}}{E_{\vec{p}}}} e^{i\vec{p}'\cdot\vec{x}} (2\pi)^3 \delta^3(\vec{p}-\vec{p}')|0\rangle \\ &= e^{i\vec{p}\cdot\vec{x}}\end{aligned}$$

$\rightarrow$ the position space rep. of the single particle wavefunction, just as $\langle x|\vec{p}\rangle \sim e^{i\vec{p}\cdot\vec{x}}$ is the w.f. of $|\vec{p}\rangle$

The Heisenberg Picture

Thus far we have taken the field operators to be static, assuming the states evolve in time (w/ evolution governed by the Hamiltonian)

Now we'll move to the opposite picture, in which the field operators (and all the others) vary in time. The advantage we'll see is that the ~~Heisenberg~~ Heisenberg field satisfies the Klein-Gordon equation, getting us closer to a ~~Lorentz~~ Lorentz-invariant picture. We'll also be able to study notions of causality.

From here, we'll ~~be~~ turn to Green's functions of the Klein-Gordon operator — ~~the~~ one such function, the Feynman propagator — will be an important ingredient in the Feynman rules we come back to soon.

The Heisenberg operators are constructed via the relation:

$$\underbrace{\phi(x)}_{\text{Heisenberg field}} \equiv \phi(t, x) = e^{iHt} \underbrace{\phi(\vec{x})}_{\text{Schrödinger field}} e^{-iHt}$$

and similarly for $\pi(x) = \pi(t, x)$.

The Heisenberg equation of motion for an operator O is:

$$i \frac{\partial}{\partial t} O = [O, H]$$

We can apply this to $\phi(t, \vec{x})$:

$$i \frac{\partial}{\partial t} \phi(t, \vec{x}) = \left[\phi(t, \vec{x}), \int d^3x' \left\{ \frac{1}{2} \dot{u}^2(t, x') + \frac{1}{2} (\vec{\nabla} \phi(t, \vec{x}'))^2 + \frac{1}{2} m^2 (\phi(t, x'))^2 \right\} \right]$$

The commutators of the last two terms vanish, so we care only about

$$[\phi(t, x), u^2(t, x')]$$

$$\begin{aligned} \text{The general identity is } [A, B^2] &= ABB - BBA \\ &= ABB - BAB + BAB - BBA \\ &= [A, B]B + B[A, B] \end{aligned}$$

$$\text{w/ } [\phi(t, \vec{x}), u(t, \vec{x}')] = i \delta^3(\vec{x} - \vec{x}')$$

(at equal times)

$$\begin{aligned} \Rightarrow i \frac{\partial}{\partial t} \phi(t, x) &= \int d^3x' i \delta^3(\vec{x} - \vec{x}') u(t, \vec{x}) \\ &= i u(t, \vec{x}) \quad (\text{trivially}) \end{aligned}$$

One could also find the Heisenberg equation of motion for $\dot{u}(t, x)$: similar steps yield:

$$i \frac{\partial}{\partial t} \dot{u}(t, \vec{x}) = -i (\vec{\nabla}^2 + m^2) \phi(t, \vec{x})$$

Combining these results:

$$i \frac{\partial}{\partial t} \hat{u}(t, x) = i \frac{\partial}{\partial t} \frac{\partial}{\partial t} \phi(t, x) = i (\vec{\nabla}^2 - m^2) \phi(t, x)$$

$$\Rightarrow \frac{\partial^2}{\partial t^2} \phi(t, x) = (\vec{\nabla}^2 - m^2) \phi(t, x)$$

→ the Klein-Gordon equation for a field operator, as promised.

It's useful to unpack the time-dependence in terms of the ladder operators. First, note:

$$H a_{\vec{p}} = a_{\vec{p}} (H - E_{\vec{p}})$$

$$\Rightarrow H^n a_{\vec{p}} = a_{\vec{p}} (H - E_{\vec{p}})^n$$

Similarly,

$$H a_{\vec{p}}^\dagger = a_{\vec{p}}^\dagger (H + E_{\vec{p}}) \Rightarrow H^n a_{\vec{p}}^\dagger = a_{\vec{p}}^\dagger (H + E_{\vec{p}})^n$$

From these, we can show:

$$e^{iHt} a_{\vec{p}} e^{-iHt} = a_{\vec{p}} e^{-iE_{\vec{p}}t}$$

$$e^{iHt} a_{\vec{p}}^\dagger e^{-iHt} = a_{\vec{p}}^\dagger e^{+iE_{\vec{p}}t}$$

(These are easily proven w/ $e^{iHt} = \sum_n \frac{(it)^n H^n}{n!}$)

Now we can use these in the mode expansions. Starting from $\phi(\vec{x})$:

$$\phi(x) = \int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} (a_{\vec{p}} e^{i\vec{p}\cdot\vec{x}} + a_{\vec{p}}^\dagger e^{-i\vec{p}\cdot\vec{x}})$$

$$\rightarrow \phi(t, \vec{x}) = e^{iHt} \phi(\vec{x}) e^{-iHt}$$

$$= \int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \left(e^{iHt} a_{\vec{p}} e^{-iHt} e^{i\vec{p}\cdot\vec{x}} + e^{iHt} a_{\vec{p}}^\dagger e^{-iHt} e^{-i\vec{p}\cdot\vec{x}} \right)$$

$$= \int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \left(a_{\vec{p}} e^{-iE_{\vec{p}}t + i\vec{p}\cdot\vec{x}} + a_{\vec{p}}^\dagger e^{+iE_{\vec{p}}t - i\vec{p}\cdot\vec{x}} \right)$$

$$= \int \frac{d^3 \vec{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \left(a_{\vec{p}} e^{-ip\cdot x} + a_{\vec{p}}^\dagger e^{+ip\cdot x} \right) \Big|_{p^0 = E_{\vec{p}}}$$

where $p \cdot x = p^0 x^0 - \vec{p} \cdot \vec{x}$. Also, $\hat{u}(t, \vec{x}) = \frac{\partial}{\partial t} \phi(t, \vec{x})$

This already looks more Lorentz covariant.

We can take this further by noting that the above operations can be performed not only w/ H , but w/ $\vec{P}$ - the momentum operator.

In particular,

$$e^{-i\vec{P}\cdot\vec{x}} a_{\vec{p}} e^{i\vec{P}\cdot\vec{x}} = a_{\vec{p}} e^{i\vec{p}\cdot\vec{x}}$$

$$e^{-i\vec{P}\cdot\vec{x}} a_{\vec{p}}^{\dagger} e^{i\vec{P}\cdot\vec{x}} = a_{\vec{p}}^{\dagger} e^{-i\vec{p}\cdot\vec{x}}$$

and therefore:

$$\begin{aligned}\phi(x) = \phi(t, \vec{x}) &= e^{i(Ht - \vec{P}\cdot\vec{x})} \phi(0, 0) e^{-i(Ht - \vec{P}\cdot\vec{x})} \\ &= e^{i\vec{P}\cdot\vec{x}} \phi(0) e^{-i\vec{P}\cdot\vec{x}}\end{aligned}$$

w/ $P^{\mu} = (H, \vec{P})$ the four-momentum operator.

We can gain more insight in the physical interpretation of the field operator from the Heisenberg mode expansion:

$$\phi(x) = \int \frac{d^3\vec{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \left(a_{\vec{p}} e^{-ip\cdot x} + a_{\vec{p}}^{\dagger} e^{+ip\cdot x} \right) \Big|_{p^0 = E_{\vec{p}}}$$

→ operator which creates/destroys particles, which are quanta of the field excitation

→ also, linear combination of solutions ($e^{ip\cdot x}$, $e^{-ip\cdot x}$) to the Klein-Gordon equation.

Note: both signs of time-dependence appear:

$e^{-ip^0 t}$, $e^{+ip^0 t}$ → but $E_{\vec{p}} = p^0$ is always positive.

In ordinary QM, two signs would correspond to positive & negative energies.

In this context of fields, though, the interpretation is different — let's refer to them as positive & negative frequency modes.

A positive frequency ($e^{-ip \cdot x}$) solution has a $-$ annihilation $-$ operator as its coefficient, which destroys a particle in the single-particle wavefunction.

A negative frequency solution ($e^{ip \cdot x}$) is the Hermitian conjugate, and so has the creation operator, which creates a particle in the positive-energy single particle wavefunction.

→ Relativistic wave equations have $+$ & $-$ frequency solutions, but consistent w/ requirement that the quantum theory has only positive energy excitations.

We haven't discussed particles vs. anti-particles here — that's because we're dealing w/ a real field, which is its own antiparticle.

We've seen that if we have a complex field, it naturally can have a (global) $U(1)$ charge — then what?

In a theory of complex fields, we are led to consider two distinct fields: ϕ and $\phi^\dagger$.

The field $\phi(x)$ will create positively charged particles, and destroy negatively charged antiparticles, while $\phi^\dagger(x)$ creates negatively charged antiparticles & destroys positively charged particles.

→ in both cases, we have positive/negative frequency modes, but only positive energy excitations.

Causality

Now we want to see that our relativistic theory is causal. In general, this requires that all spacelike operators commute:

$$[O_1(x), O_2(y)] = 0 \text{ if } (x-y)^2 < 0$$

→ then, a prior measurement of O_1 cannot affect a measurement of O_2 , & vice-versa.

So, consider $\phi(x)$, $\phi(y)$ at spacetime points x^μ, y^μ : Define:

$$\Delta(x-y) = [\phi(x), \phi(y)]$$

If this vanishes at x around some neighborhoods of x, y then other operators built out of ϕ will respect similarly vanishing commutators.

As always, we plug in the mode expansion:

$$\begin{aligned} \Delta(x-y) &= \int \frac{d^3 p}{(2\pi)^3} \int \frac{d^3 q}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \frac{1}{\sqrt{2E_{\vec{q}}}} \times \\ &\quad \left[a_{\vec{p}} e^{-ip \cdot x} + a_{\vec{p}}^\dagger e^{ip \cdot x}, a_{\vec{q}} e^{-iq \cdot y} + a_{\vec{q}}^\dagger e^{iq \cdot y} \right] \\ &= [a_{\vec{p}}, a_{\vec{q}}^\dagger] e^{-ip \cdot x + iq \cdot y} + [a_{\vec{p}}^\dagger, a_{\vec{q}}] e^{ip \cdot x - iq \cdot y} \\ &= (2\pi)^3 \delta^3(\vec{p} - \vec{q}) - (2\pi)^3 \delta^3(\vec{p} - \vec{q}) \end{aligned}$$

Performing the $d^3 q$ integral:

$$\Delta(x-y) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} \left[e^{-ip \cdot (x-y)} - e^{ip \cdot (x-y)} \right]$$

→ Lorentz invariant, so $\Delta(x-y)$ is!

For time like separations, e.g. $(x-y)^\mu = (t, 0)$ we get for large times: $t \gg 1/m$

$$\Delta(x-y) = [\phi(t, \vec{x}), \phi(0, \vec{x})] \sim e^{-imt} - e^{+imt}$$

→ this doesn't vanish, as expected, since timelike separations are causally connected.

For spacelike separations, however, we can consider

$$x^0 = y^0 \quad \wedge \quad (x-y)^2 = -(\vec{x}-\vec{y})^2 < 0$$

Then,

$$\begin{aligned} \Delta(x-y) &= [\phi(t, \vec{x}), \phi(t, \vec{y})] \\ &= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} [e^{i\vec{p} \cdot (\vec{x}-\vec{y})} - e^{-i\vec{p} \cdot (\vec{x}-\vec{y})}] \end{aligned}$$

But here we can change variables in the second term:

Under $\vec{p} \rightarrow -\vec{p}$, the $\int d^3p$ is invariant after changing the order of integration, so

$$- \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} e^{-i\vec{p} \cdot (\vec{x}-\vec{y})} \rightarrow - \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} e^{+i\vec{p} \cdot (\vec{x}-\vec{y})}$$

$$\rightarrow \Delta(x-y) = 0!$$

We found that

$$\Delta(x-y) = [\phi(t, \vec{x}), \phi(t, \vec{y})] = 0$$

$$\text{for } -(\vec{x}-\vec{y})^2 < 0$$

But we also argued that $\Delta(x-y)$ is Lorentz-invariant, so this holds for any spacelike separated point.

We thus see that causality is maintained in our quantum field theory.

Green's Functions and the Feynman Propagator

We now wish to study Green's functions of the Klein-Gordon operator, i.e., functions satisfying the differential equation

$$(\square + m^2) G(x-y) = -i \delta^4(x-y)$$

(Note that we're now fully relativistic!)

Taking Fourier transforms:

$$G(x-y) = \int \frac{d^4 p}{(2\pi)^4} \tilde{G}(p) e^{-ip \cdot (x-y)}$$

$$\delta^4(x-y) = \int \frac{d^4 p}{(2\pi)^4} e^{-ip \cdot (x-y)}$$

We find:

$$(-p^2 + m^2) \hat{G}(p) = -i$$

$$\text{So } \hat{G}(p) = \frac{i}{p^2 - m^2} \quad \left| \quad G(x-y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2} e^{-ip(x-y)} \right|$$

This is the solution, but it's not completely well-defined, as ~~it~~ it includes integrating over singularities at

$$p^2 - m^2 = 0, \text{ or } p^2 = m^2, \text{ or } p^0 = \pm E_{\vec{p}}$$

We have to regulate the integral at these poles. There are several ways to do this, but the answer we want is known as the Feynman propagator, which will be denoted $D_F(x-y)$.

To arrive at $D_F(x-y)$, we introduce the "iε prescription", writing:

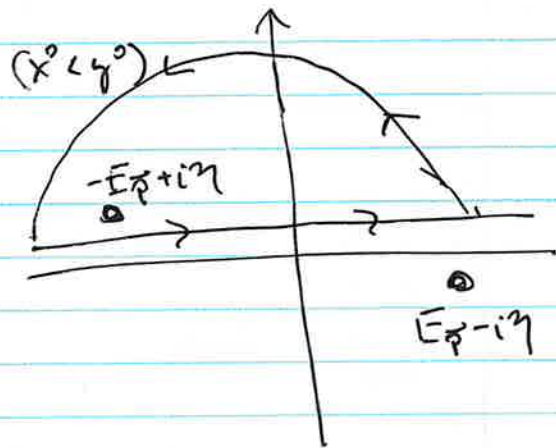
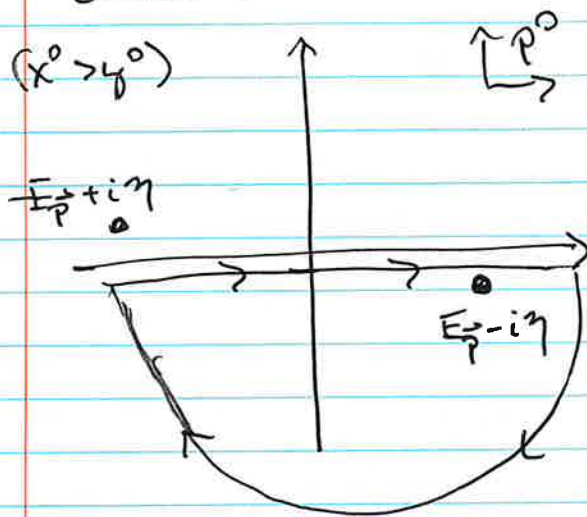
$$D_F(x-y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{-ip(x-y)}$$

Now the poles are at $p^0 = \pm (E_{\vec{p}} - i\eta)$, w/

$$\eta = \epsilon / 2E_{\vec{p}}$$

→ The pole ~~near~~ the positive E_F is displaced below the real axis, while the pole near $-E_F$ is displaced above.

We can consider this a contour integral in the complex p^0 plane. For $x^0 > y^0$ and $x^0 < y^0$, we take different choices.



Now the integral over the semicircle vanishes in each case, due to the damping term.

$$e^{-ip^0(x^0 - y^0)} > e^{-|\text{Im}' p^0|(x^0 - y^0)}, \quad x^0 > y^0$$

$$e^{-ip^0(x^0 - y^0)} > e^{-|\text{Im}' p^0|(y^0 - x^0)}, \quad x^0 < y^0$$

In each case we pick up one pole in the contour, and apply the residue theorem. For $x^0 > y^0$,

$$\oint \frac{dp^0}{2\pi} \frac{i}{(p^0 - E_F)(p^0 + E_F)} e^{-ip^0(x^0 - y^0)} = -2\pi i \text{Res}(p^0 = \pm E_F)$$

$$= \frac{+1}{2E_F} e^{-iE_F(x^0 - y^0)}$$

So,

$$D_F(x-y) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} e^{-ip \cdot (x-y)} \Big|_{p^0 = +E_{\vec{p}}} \quad (x^0 > y^0)$$

A similar analysis for $x^0 < y^0$ yields:

$$D_F(x-y) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} e^{-ip \cdot (y-x)} \Big|_{p^0 = +E_{\vec{p}}} \quad (x^0 < y^0)$$

$\vec{p}$
 opposite spatial
 order!

Now we can define the two-point correlation function:

$$D(x-y) = \langle 0 | \phi(x) \phi(y) | 0 \rangle$$

Some algebra with the mode expansions yields:

$$D(x-y) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} e^{-ip \cdot (x-y)}$$

So we can write:

$$D_F(x-y) = \begin{cases} D(x-y) & \text{for } x^0 > y^0 \\ D(y-x) & \text{for } y^0 > x^0 \end{cases}$$

This is often summarized as:

$$D_F(x-y) = \theta(x^0 - y^0) \langle 0 | \phi(x) \phi(y) | 0 \rangle \\ + \theta(y^0 - x^0) \langle 0 | \phi(y) \phi(x) | 0 \rangle \\ \equiv \langle 0 | T \phi(x) \phi(y) | 0 \rangle$$

Where " T " denotes the time-ordering operation — all the operators after T are ordered w/ the latest in time to the left.

This solution will be used repeatedly when we start evaluating scattering amplitudes.

With that, we have essentially solved the free theory — we know the Hilbert space, how fields are correlated over spacetime; there's not much left to solve.

Next we'll mix things up by adding interactions.